Department of Information Technology
Academic Year 2025 - 2026 (Even Semester)

Semester, Degree & Branch: VI Semester, B.Tech. IT
Course Code & Title: CCS354 & Network Security
Name of the Faculty member (s): Mrs. G.Sivasathiya, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit I / Message Authentication
- **Course Outcome:** CO1
- **Topic Learning Outcome: TLO3**
- **Activity Chosen:** Flash Cards

- **Justification:**

  Message authentication ensures message integrity, data origin authenticity, and sometimes non-repudiation, defending against modification, replay, and via digital signatures, source spoofing. In flash cards activity the students will be able to focus on essential information rather than every minor detail. With this method we can ensure the quality of student's comprehension by reading both the prompt and the answer.

- **Time Allotted for the Activity:** 15 Minutes

- **Details of the Implementation:**
  - At the end of the lecture, students were grouped as 5 or 6 per team.
  - One student in the group has been provided with 3 minimum flash cards that requires filling in missing words, or giving one-line definition for the given question / topic.
  - Fig 1.a and Fig 1.b shows the sample flash cards and its respective answers created. Fig. 1 c shows G. Abinaya is asking questions present in the flash card and getting answers from her peer students.
  - Team with maximum correct answers was announced as winners. This activity demonstrated the involvement of students in the particular class.

- **CO – PO / PSO mapping:**

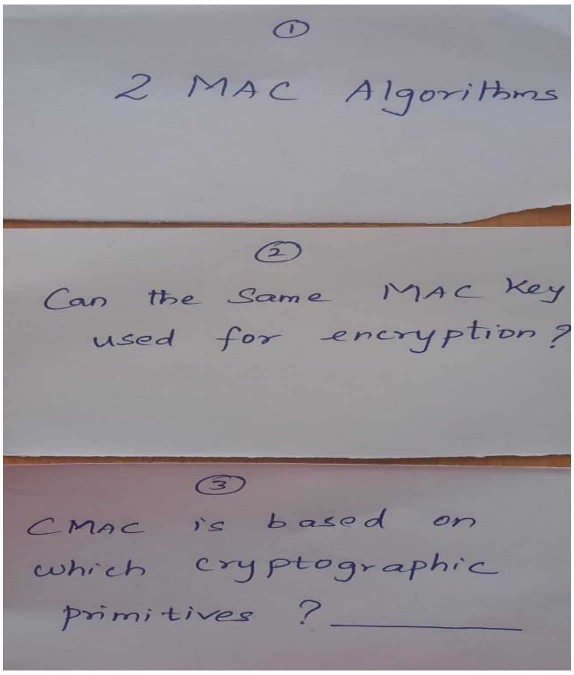
CO	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	2	2	2	2	1	2	1	2	3	1

(1 – Low 2 – Moderate 3 – High)

PO/PSO Mapped:

Innovative Practice	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	2	2	2	2	1	2	1	2	3	1
Justification for Correlation	Understanding cryptography concepts requires a solid foundation in engineering principles to grasp the technical aspects of security protocols, threat analysis, and protective measures.	Analyzing the need for encryption techniques involves identifying potential threats and vulnerabilities in network systems, which is essential for effective risk management and protection	Recognizing the need for cryptography involves considering security measures and algorithm during the design and development phases to ensure the delivery of safe and resilient solutions.	Cryptography often involves analyzing security algorithms and identifying potential weaknesses, which requires analytical skills and problem-solving abilities	Understanding the need for cryptography involves familiarity with modern encryption algorithms and tools to secure data and enhance privacy.	Effective cryptography requires collaboration within teams and individual contributions to design and implement encryption algorithms, ensuring robust data security and privacy	Communicating the significance of cryptography and its impact on data security to stakeholders is essential for ensuring alignment and garnering support for encryption initiatives	Understanding the need for cryptography involves considering its implications on project timelines and budgets, necessitating effective project management and financial planning for secure system development	Recognizing the dynamic nature of cryptography, engineers should cultivate a mindset of continuous learning to stay updated with evolving encryption techniques and security technologies	Apply the knowledge of proficiency in cryptography and provide the foundation to ensuring robust encryption of complex computational problems.	Apply the knowledge of cryptography to provide solutions to emerging technologies such as mobile application security, ensuring the protection of sensitive data and providing reliable IT solutions	Developing cryptographic solutions is a base for implementing a project as an ethical software engineer, ensuring the secure handling of data and protecting user privacy .

- **Images / Screenshot of the practice:**



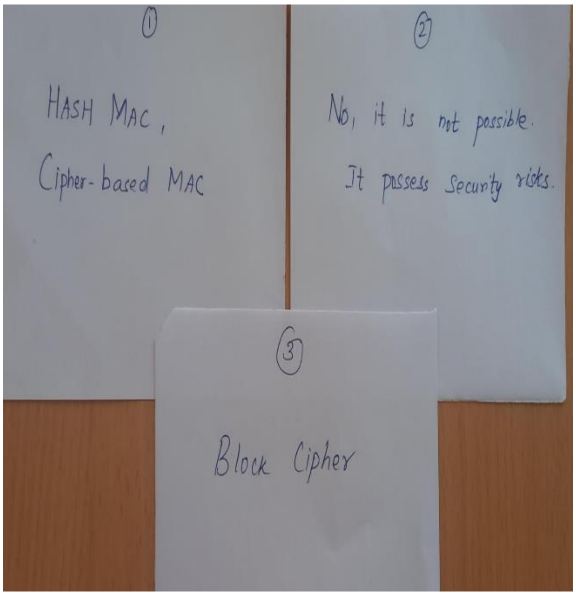


Figure 1. a Flash card – Questions

Figure 1. a Flash card - Answers

Figure 1. c G.Abinaya is having questionnaire session to her peer-mates with the given flashcards

Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***
  - Students reported that interactive, visually appealing flashcards made learning more enjoyable and less monotonous.
  - Educators reported that incorporating local wisdom and visuals helps bridge abstract concepts with students' real-life experiences.
- ***Benefit of the practice:***
  - Students were actively participated in this activity.
  - This activity forces the brain to retrieve information, strengthening memory pathways more effectively than passive reading.
  - It enhances focus, concentration, and rapid processing in students by engaging both sides of the brain.
  - It Strengthens language skills through visual stimulation and image-word association
- ***Challenges faced in implementation:***
  - They are best for reinforcing known information, not for learning complex subjects from scratch, leading to frustration if used inappropriately.
  - Information on cards often lacks real-world context, making it difficult for learners to apply the knowledge practically.
- ***References:***

  - https://fliphtml5.com/learning-center/flashcard-design-ideas/
  - https://www.kutuki.in/post/10-ways-to-make-learning-fun-with-flashcards
  - https://usm.maine.edu/learning-commons/using-flash-cards/
  - https://www.bookwidgets.com/blog/2022/10/20-amazing-ways-to-use-digital-flashcards-in-your-classroom

Signature of Faculty Member **HOD**